

**Exponents and Scientific Notation**

41 points

Name \_\_\_\_\_

Date \_\_\_\_\_

Period \_\_\_\_\_

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*Show your work. Circle your final answer.*

**This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.**

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**1. Laws of Exponents**9 points

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1. Write each expression as a single power.

a.  $4^3 \cdot 4^5$  1 pt

b.  $(2^3)^4$  1 pt

c.  $\frac{7^9}{7^4}$  1 pt

d.  $9^0$  (give its value) 1 pt

2. Find the value of each expression.

a.  $\left(\frac{2}{3}\right)^3$

1 pt

b.  $6^0 \cdot 5^2$

1 pt

c.  $\frac{(10^2)^2}{10^3}$

1 pt

3. Which expression is equivalent to  $2^4 \cdot 2^3$  ?

1 pt

- ☐ A.  $4^7$
- ☐ B.  $2^{12}$
- ☐ C.  $4^{12}$
- ☐ D.  $2^7$

4. Find the value of  $n$  that makes the statement true:  $\frac{5^n}{5^2} = 125$

1 pt

## 2. Integer Exponents

11 points

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5. Rewrite each expression with positive exponents. Do not evaluate.

a.  $6^{-3}$  1 pt

b.  $\frac{1}{(-2)^{-5}}$  1 pt

c.  $3 \cdot 10^{-4}$  1 pt

6. Find the value of each expression.

a.  $4^{-2}$  1 pt

b.  $(-3)^{-3}$  1 pt

c.  $\left(\frac{1}{2}\right)^{-3}$  1 pt

d.  $2^5 \cdot 2^{-7}$  1 pt

7. Which is the value of  $5^{-2}$  ?

1 pt

- ☐ A.  $\frac{1}{10}$
- ☐ B.  $-25$
- ☐ C.  $\frac{1}{25}$
- ☐ D.  $-10$

8. Select ALL of the expressions whose value is  $\frac{1}{64}$  .

1 pt

- ☐ A.  $2^{-6}$
- ☐ B.  $4^{-3}$
- ☐ C.  $-8^2$
- ☐ D.  $2^{-8}$
- ☐ E.  $\left(\frac{1}{4}\right)^3$

9. Marco says the value of  $3^{-2}$  is  $-9$  . Explain why he is wrong, and give the correct value. 1 pt

10. Which is greater,  $2^{-3}$  or  $3^{-2}$  ? Show how you know.

1 pt

### 3. Variable Bases

9 points

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11. Write each expression as a single monomial.

a.  $x^4 \cdot x^6$  1 pt

b.  $(y^3)^5$  1 pt

c.  $(3a^2)^3$  1 pt

d.  $\frac{15m^7}{5m^2}$  1 pt

12. Rewrite each expression using only positive exponents.

a.  $b^{-4}$  1 pt

b.  $5x^{-2}$  1 pt

c.  $(2m^{-3}n)^2$  1 pt

13. Which expression is equivalent to  $(4x^3)^2$  ?

1 pt

- ☐ A.  $16x^5$
- ☐ B.  $8x^6$
- ☐ C.  $4x^6$
- ☐ D.  $16x^6$

14. How are  $x^2$  and  $x^{-2}$  related? Explain, and give the value of each when  $x = 5$  .

1 pt

#### 4. Scientific Notation

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12 points

15. Write each number in scientific notation.

a. 8,300,000

1 pt

b. 0.00047

1 pt

c. 60,500,000,000

1 pt

d. 0.0000000092

1 pt

16. Write each number in standard form.

a.  $2.9 \times 10^5$  1 pt

b.  $7.1 \times 10^{-6}$  1 pt

17. Which is 0.0036 written in scientific notation? 1 pt

- ☐ A.  $3.6 \times 10^3$
- ☐ B.  $3.6 \times 10^{-3}$
- ☐ C.  $3.6 \times 10^{-2}$
- ☐ D.  $36 \times 10^{-4}$

18. Answer each question.

a. How many times larger is  $9 \times 10^9$  than  $3 \times 10^6$  ? 1 pt

b. How many times larger is  $4.8 \times 10^{-3}$  than  $1.6 \times 10^{-7}$  ? 1 pt

19. Which is greater,  $8.9 \times 10^{-4}$  or  $2.1 \times 10^{-3}$  ? Explain how you know. 1 pt

**20.** A computer displays a number as 5.6E8. Write the number in standard form.

1 pt

**21.** Between which two consecutive powers of 10 does  $4.7 \times 10^{-5}$  lie?

1 pt